Homework 5

Due: Thursday, March 5, 2026 at 12:00pm **(Noon)**

Project Report

No-pruning Training data loss: 0.0
No-pruning Test data loss: 0.04933110367892977
Pruning Training data loss: 0.022
Pruning Testing loss: 0.039297658862876256
Building a pruned tree successfully!
No-pruning Training data loss: 0.0
No-pruning Test data loss: 0.014214046822742474
Pruning Training data loss: 0.004
Pruning Testing loss: 0.013377926421404682
Building a pruned tree successfully!
No-pruning Training data loss: 0.0
No-pruning Test data loss: 0.014214046822742474
Pruning Training data loss: 0.004
Pruning Testing loss: 0.013377926421404682
Building a pruned tree successfully!
No-pruning Training data loss: 0.016
No-pruning Test data loss: 0.1503267973856209
Pruning Training data loss: 0.062
Pruning Testing loss: 0.11610918877354863
Building a pruned tree successfully!
No-pruning Training data loss: 0.014
No-pruning Test data loss: 0.12572087658592848
Pruning Training data loss: 0.043
Pruning Testing loss: 0.1249519415609381
Building a pruned tree successfully!
No-pruning Training data loss: 0.014
No-pruning Test data loss: 0.13494809688581316
Pruning Training data loss: 0.043
Pruning Testing loss: 0.12648981161091888
Building a pruned tree successfully!

(a). Comment on the results of your final program. Discuss the differences in training and test error of pruned and non-pruned trees. Which measure of gain most effectively reduced training error? Was pruning effective? Use tables or graphs to demonstrate your findings. **(5 points)**

Dataset	Gain Function	Train (unpruned)	Test (unpruned)	Train (pruned)	Test (pruned)
Chess	Error	0.0000	0.0493	0.0220	0.0393
Chess	Entropy	0.0000	0.0142	0.0040	0.0134
Chess	Gini	0.0000	0.0142	0.0040	0.0134
Spam	Error	0.0160	0.1503	0.0620	0.1161
Spam	Entropy	0.0140	0.1257	0.0430	0.1250
Spam	Gini	0.0140	0.1349	0.0430	0.1265

On the Chess dataset, all three gain functions produced unpruned trees with perfect training loss (0.0), indicating good fit of the training set. Entropy and Gini had lower test loss (0.0142) compared to

misclassification error (0.0493), so they produce more generalizable splits. On the Spam dataset, no gain function achieved zero training loss, and test loss was considerably higher across the board, indicating this task is harder.

Pruning consistently increased training loss, and reducing test loss in most cases. The reduction in test loss was most noticeable on Chess and moderate on Spam. Overall, **entropy and Gini most effectively minimized both training and test error**, performing identically on both datasets and outperforming misclassification error as a gain function. Pruning was effective in all cases.

(b). Using the `spam.csv` dataset, plot the loss of your decision tree on the *training* set for trees with maximum depth set to each value between 1 to 15. For these plots, the trees should not be pruned and you can use the entropy gain function. Discuss any trends you find and attempt to explain them in three to five sentences. **(10 points)**

(c). Read the short excerpt below from a paper written by Rachel L. Thomas and David Uminsky where they bring up the issue of Machine Learning's dependence on metrics:

Goodhart's Law, that a measure that becomes a target ceases to be a good measure, remains a relevant lens to see the shortcomings of ML's reliance on metrics. By definition, machine learning is a process in which a measure is the target. Thus, there is an increased likelihood that any risks of optimizing metrics are heightened by AI (Slee 2019), and Goodhart's Law grows increasingly relevant. For instance, if an algorithm learns to produce answers that humans find appealing as opposed to accurate answers.'

What are your initial thoughts on reading this passage? Do you agree with the author? Why or why not? Please elaborate your answer.
I mostly agree. Goodhart's Law is a great description of a failure mode that shows up alot in ML.The main issue is that ML systems are optimized for their metric, making them the most aggressive possible breakers of Goodhart's Law. A model trained to maximize accuracy on a benchmark doesn't care about the overall goal, it finds the fastest path to a high number, which may diverge from what you want. The RLHF example in the passage is a good real case of this. That said, the solution isn't to abandon metrics, since without them Ml doesn't work. The authors point is better read as a call for many chosen metrics, held rather than blind trust in any single loss function or benchmark score. The field has responded somewhat with things like held-out test sets, red-teaming, and diverse eval suites, though these are imperfect patches on a deep structural issue. the authors are right that this risk is real especially as models become powerful enough to work around sophisticated metrics.